

```

# General Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer

import warnings
import gspread
from google.colab import drive
from gspread_dataframe import set_with_dataframe
from google.colab import auth
from google.auth import default
from oauth2client.service_account import ServiceAccountCredentials
import pandas as pd

# Configs:
warnings.filterwarnings('ignore')
drive.mount('/content/drive')

# Google Authentication:
auth.authenticate_user()
creds, _ = default()
gc = gspread.authorize(creds)

# Visualisation
import matplotlib.pyplot as plt
import seaborn as sns

# Global constants
TIME_FMT = '%H:%M'
DATE_FMT = '%d-%m-%Y'
SEED = 359
# load data
df = pd.read_excel('/content/drive/My Drive/Colab Notebooks/Zomato Dataset.xlsx')

# Kiểm tra data gốc
print("=== 5 DÒNG ĐẦU ===")
print(df.head())

print("\n=== THỐNG KÊ TIME_TAKEN ===")
print(df['Time_taken (min)'].describe())

# Vẽ phân phối thời gian
plt.figure(figsize=(10, 6))
sns.histplot(df['Time_taken (min)'], bins=30, kde=True)
plt.title("PHÂN PHỐI THỜI GIAN GIAO HÀNG")
plt.show()

# -----
# BƯỚC 3: TIỀN XỬ LÝ DATA
# -----
# Xử lý missing values (nếu có)
df.dropna(inplace=True)

# Chia features (X) và target (y)
X = df[['Weather_conditions', 'Road_traffic_density', 'Type_of_vehicle',
        'Vehicle_condition', 'multiple_deliveries', 'Festival', 'City']]
y = df['Time_taken (min)']

# Chuyển categorical features thành số
preprocessor = ColumnTransformer(
    transformers=[
        ('cat', OneHotEncoder(), ['Weather_conditions', 'Road_traffic_density',
                                  'Type_of_vehicle', 'Festival', 'City'])
    ],
    remainder='passthrough' # Giữ nguyên cột số
)

X_processed = preprocessor.fit_transform(X)

# -----
# BƯỚC 4: CHIA DATA VÀ TRAIN MODEL
# -----
X_train, X_test, y_train, y_test = train_test_split(X_processed, y, test_size=0.2, random_state=42)

# Sử dụng Random Forest (có thể điều chỉnh hyperparameter)

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model = RandomForestRegressor(n_estimators=200, max_depth=10, random_state=42)
model.fit(X_train, y_train)

# -----
# BƯỚC 5: ĐÁNH GIÁ MODEL
# -----
y_pred = model.predict(X_test)

print("\n=== ĐÁNH GIÁ MODEL ===")
print("MAE (Phút):", mean_absolute_error(y_test, y_pred))
print("RMSE (Phút):", np.sqrt(mean_squared_error(y_test, y_pred)))

# Vẽ biểu đồ so sánh thực tế vs dự đoán
plt.figure(figsize=(10, 6))
plt.scatter(y_test, y_pred, alpha=0.5)
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--')
plt.xlabel("Thực tế (phút)")
plt.ylabel("Dự đoán (phút)")
plt.title("SO SÁNH THỰC TẾ VÀ DỰ ĐOÁN")
plt.show()

# -----
# BƯỚC 6: PHÂN TÍCH PIVOT TABLE
# -----
# Tạo pivot table để kiểm tra
pivot_table = pd.pivot_table(
    df,
    values='Time_taken (min)',
    index=['Road_traffic_density', 'Weather_conditions'],
    columns=['Type_of_vehicle'],
    aggfunc=np.mean
)

print("\n=== PIVOT TABLE (TRUNG BÌNH TIME_TAKEN) ===")
print(pivot_table)

# -----
# BƯỚC 7: DỰ ĐOÁN MỚI
# -----
new_data = pd.DataFrame({
    'Weather_conditions': ['Fog'],
    'Road_traffic_density': ['Jam'],
    'Type_of_vehicle': ['motorcycle'],
    'Vehicle_condition': [2],
    'multiple_deliveries': [3.0],
    'Festival': ['No'],
    'City': ['Metropolitian']
})

new_processed = preprocessor.transform(new_data)
predicted_time = model.predict(new_processed)
print("\n=== DỰ ĐOÁN ===")
print("Thời gian giao hàng ước tính (phút):", predicted_time[0])
```

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Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
=== 5 DÒNG ĐẦU ===
  ID Delivery_person_ID Delivery_person_Age Delivery_person_Ratings \
0  0xcdcd DEHRES17DEL01 36.0 4.2
1  0xd987 KOCRES16DEL01 21.0 4.7
2  0x2784 PUNERES13DEL03 23.0 4.7
3  0xc8b6 LUDHRES15DEL02 34.0 4.3
4  0xdb64 KNPRES14DEL02 24.0 4.7

  Restaurant_latitude Restaurant_longitude Delivery_location_latitude \
0  30.327968 78.046106 30.397968
1  10.003064 76.307589 10.043064
2  18.562450 73.916619 18.652450
3  30.899584 75.809346 30.919584
4  26.463504 80.372929 26.593504

  Delivery_location_longitude Order_Date Time_Orderd Time_Order_picked \
0  78.116106 12-02-2022 21:55 22:10
1  76.347589 13-02-2022 14:55 15:05
2  74.006619 04-03-2022 17:30 17:40
3  75.829346 13-02-2022 09:20 09:30
4  80.502929 14-02-2022 19:50 20:05

  Weather_conditions Road_traffic_density Vehicle_condition Type_of_order \
0  Fog Jam 2 Snack
1  Stormy High 1 Meal
2  Sandstorms Medium 1 Drinks
3  Sandstorms Low 0 Buffet
4  Fog Jam 1 Snack

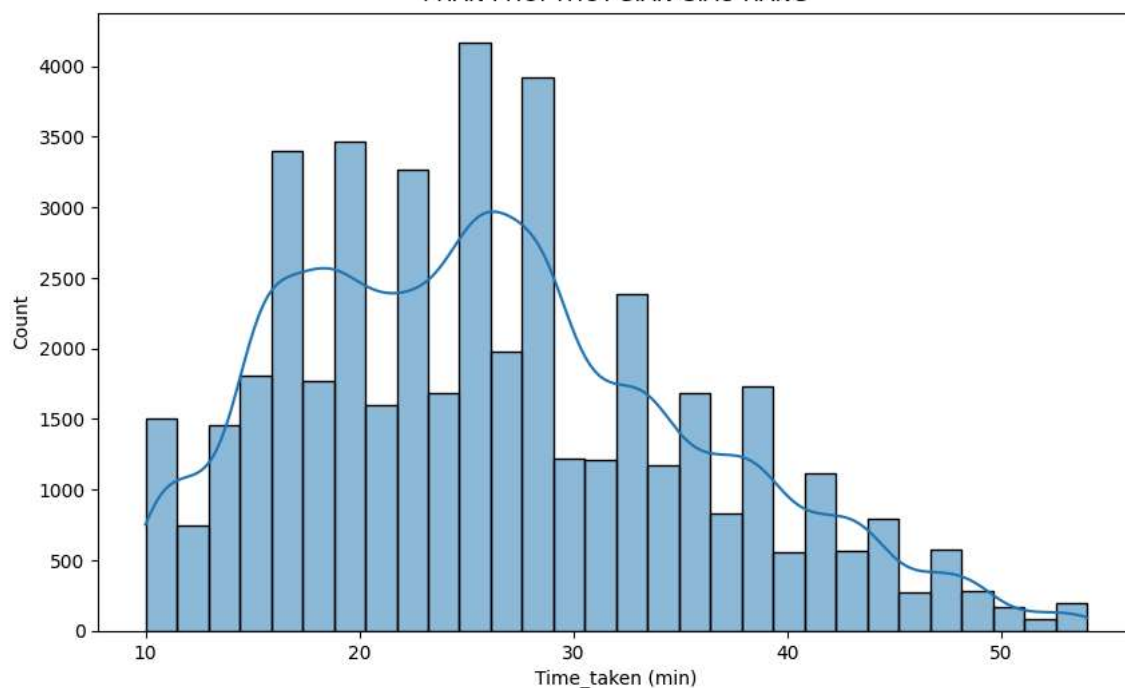
  Type_of_vehicle multiple_deliveries Festival City \
0  motorcycle 3.0 No Metropolitan
1  motorcycle 1.0 No Metropolitan
2  scooter 1.0 No Metropolitan
3  motorcycle 0.0 No Metropolitan
4  scooter 1.0 No Metropolitan

  Time_taken (min)
0  46
1  23
2  21
3  20
4  41

=== THỐNG KÊ TIME_TAKEN ===
count 45584.000000
mean 26.293963
std 9.384298
min 10.000000
25% 19.000000
50% 26.000000
75% 32.000000
max 54.000000
Name: Time_taken (min), dtype: float64

```

PHÂN PHỐI THỜI GIAN GIAO HÀNG

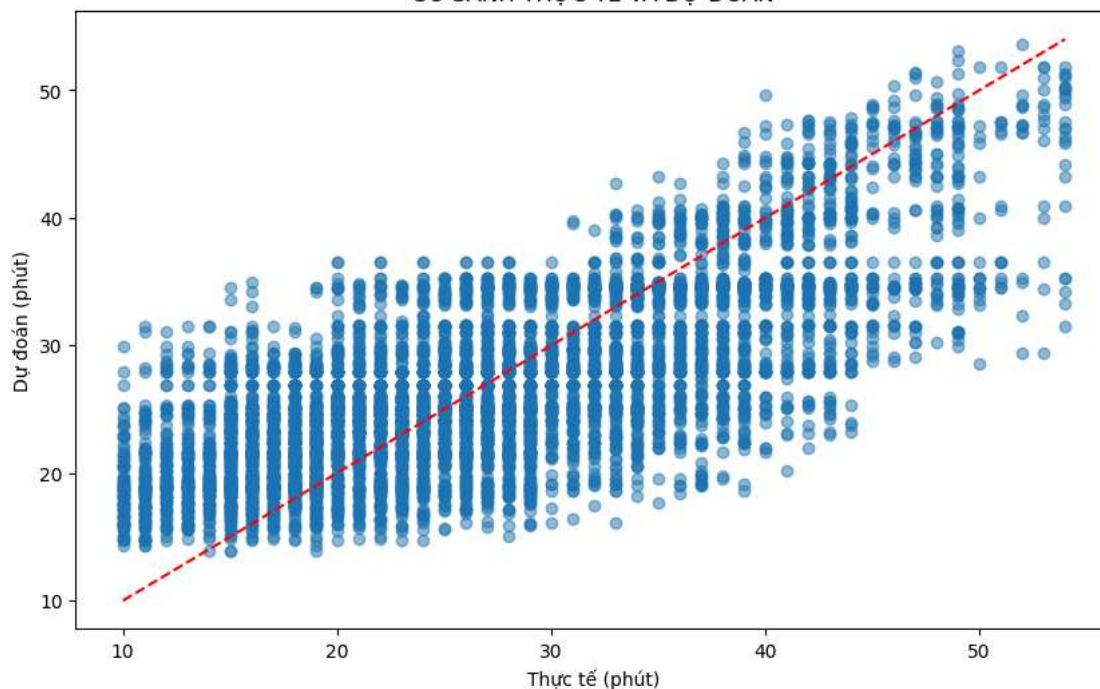


```

=== ĐÁNH GIÁ MODEL ===
MAE (Phút): 5.088049305365328
RMSE (Phút): 6.425376404741484

```

SO SÁNH THỰC TẾ VÀ DỰ ĐOÁN



=== PIVOT TABLE (TRUNG BÌNH TIME_TAKEN) ===

Type_of_vehicle		electric_scooter	motorcycle \
Road_traffic_density	Weather_conditions		
High	Cloudy	29.793103	28.475452
	Fog	28.984375	28.018779
	Sandstorms	24.190476	29.500000
	Stormy	25.264151	29.978571
	Sunny	17.607843	27.849515
	Windy	24.396226	29.380353
Jam	Cloudy	35.854430	37.125095
	Fog	36.020202	37.576664
	Sandstorms	26.568966	32.346843
	Stormy	26.288043	32.482759
	Sunny	20.052910	26.127076
	Windy	26.907104	32.685410
Low	Cloudy	20.865979	23.474892
	Fog	20.582474	23.651773
	Sandstorms	19.104396	21.444365
	Stormy	19.429319	21.676210
	Sunny	21.500000	21.776235
	Windy	19.335366	21.971059
Medium	Cloudy	26.675676	30.502075
	Fog	25.174497	30.203893
	Sandstorms	26.727941	28.874104
	Stormy	25.568182	28.767164
	Sunny	18.481818	22.009278
	Windy	26.466216	28.820116

Type_of_vehicle		scooter
Road_traffic_density	Weather_conditions	
High	Cloudy	29.811475
	Fog	29.127119
	Sandstorms	25.745370
	Stormy	24.728507
	Sunny	17.138756
	Windy	24.118421
Jam	Cloudy	36.280166
	Fog	35.884416
	Sandstorms	27.410290
	Stormy	27.297872
	Sunny	19.562232
	Windy	26.881429
Low	Cloudy	20.973350
	Fog	20.850529
	Sandstorms	19.175623
	Stormy	19.595855
	Sunny	21.666667
	Windy	19.498018
Medium	Cloudy	26.315517
	Fog	25.976628
	Sandstorms	26.283610
	Stormy	26.518966
	Sunny	18.000000
	Windy	26.646520

=== DỰ ĐOÁN ===

Thời gian giao hàng ước tính (phút): 45.804265896511914